

QUESTIONING MY SQL SERVER FAITH

... SO YOU DON'T HAVE TO



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AGENDA

- A BIT OF HISTORY
- THE CHALLENGE
- SQL SERVER FIGHTS BACK
- CONCLUSIONS

A BIT OF HISTORY

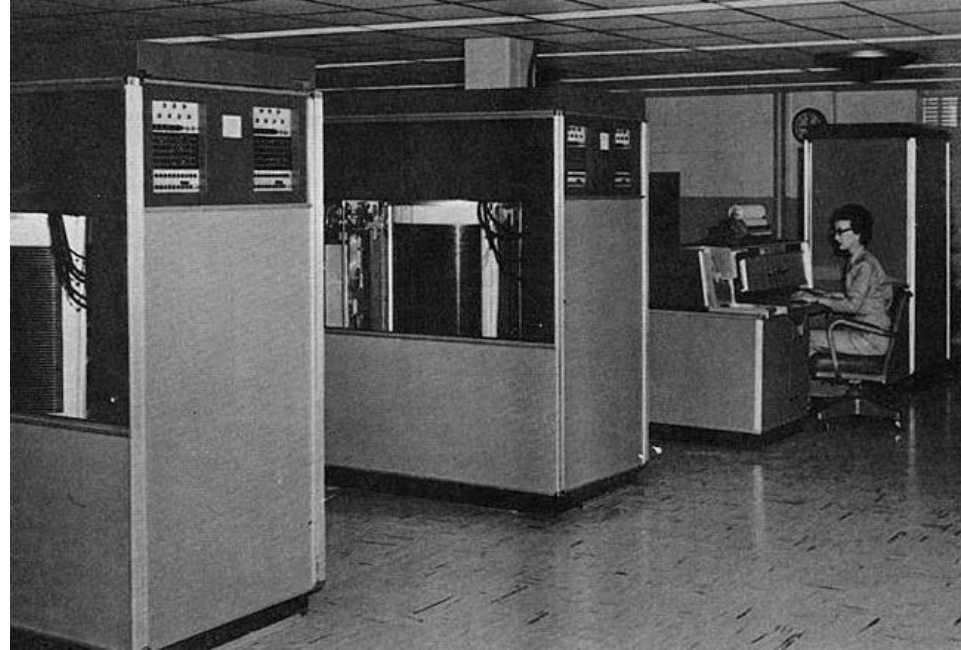




A BIT OF HISTORY

The seventies

- Early databases were ridden with inconsistencies and hard to work with
- Hardware was bulky and expensive
- Edgar Codd creates the Relational Model



1970

YOU ARE
HERE



A BIT OF HISTORY

The eighties

- Oracle dominates the scene
- Other players are slowly entering the market



1970

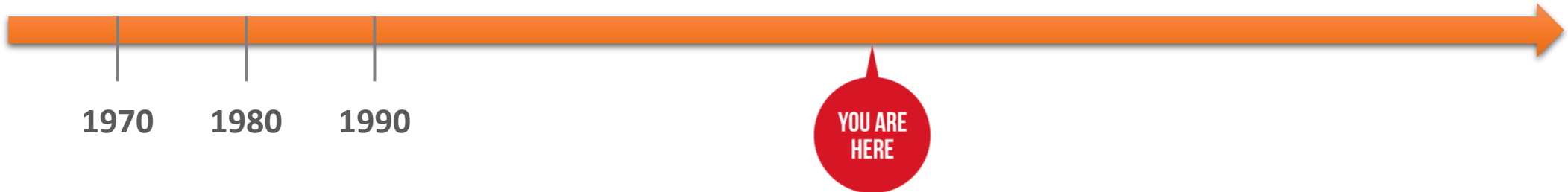
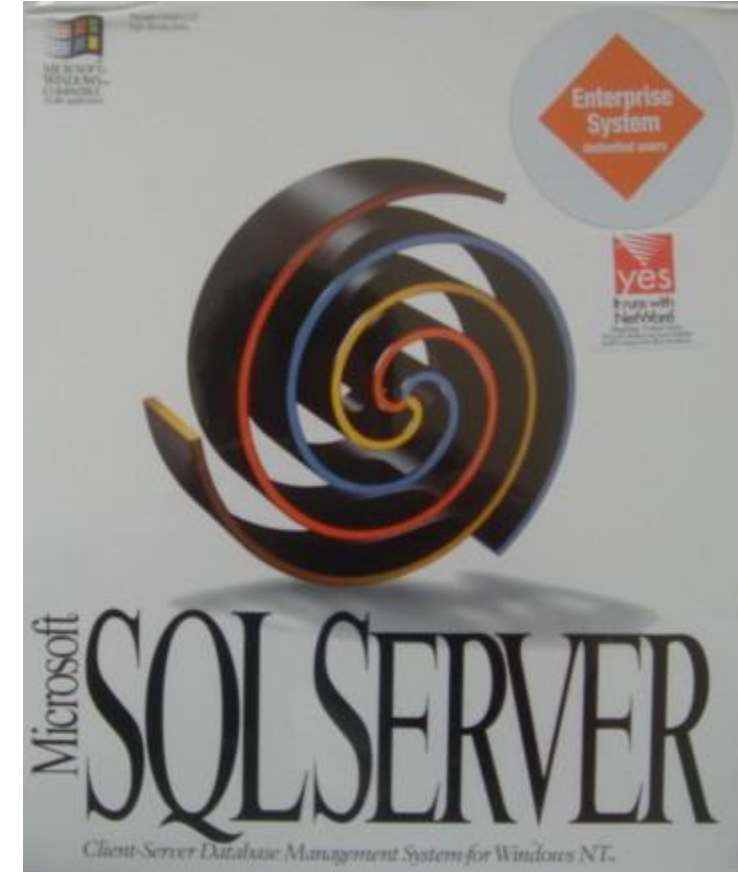
1980



A BIT OF HISTORY

The nineties

- 1989 - SQL Server 1.0
- 1992 - SQL Server 4.2 on WINNT
- 1994 - Postgres95
- 1995 - MySQL
- 1998 - SQL Server 7
- Oracle still dominates the market

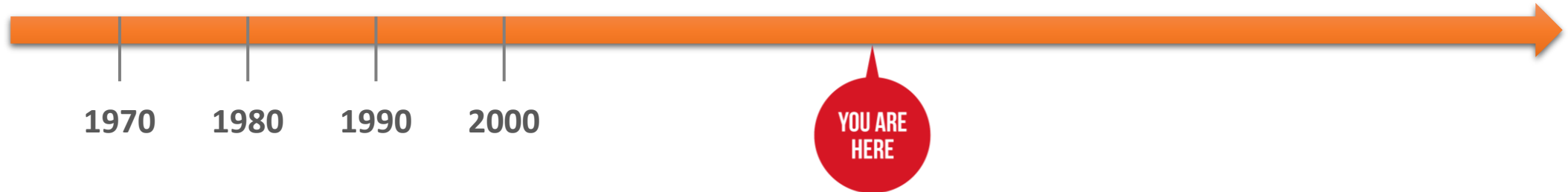
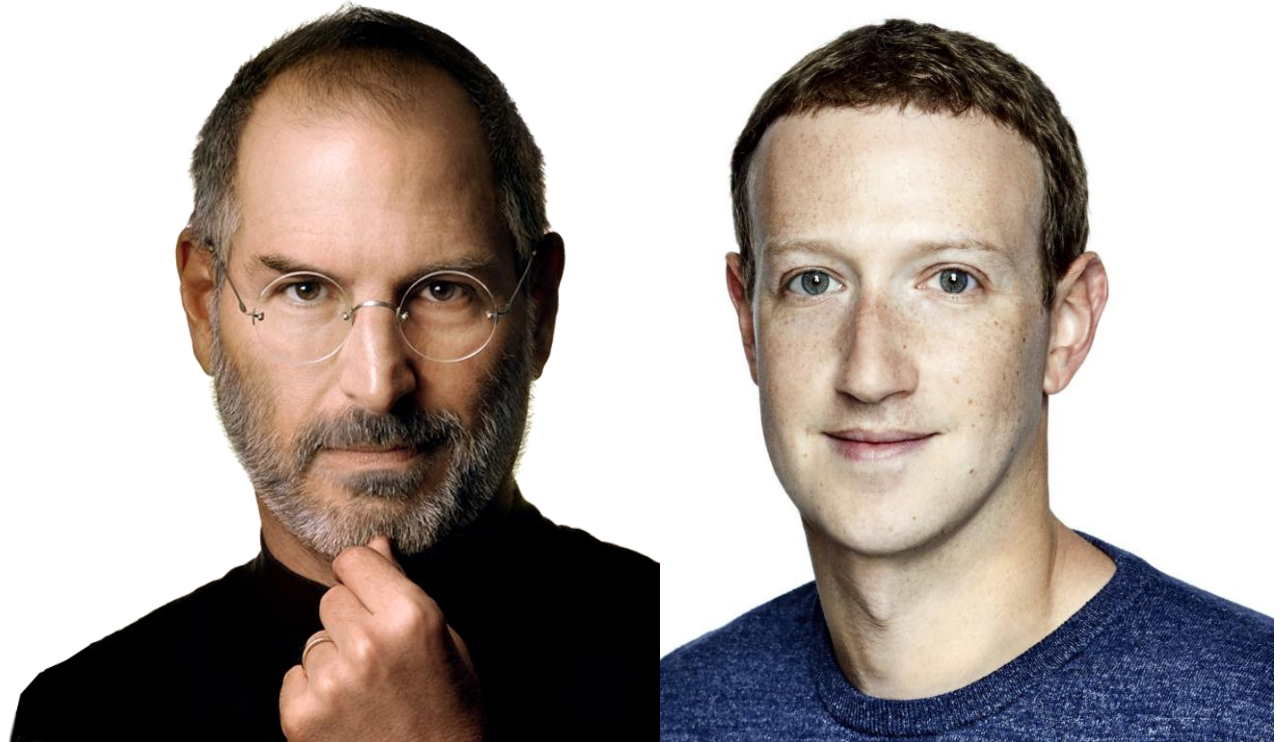




A BIT OF HISTORY

The 2000s

- 1998 – Google
- 2003 – Myspace
- 2003 – Facebook
- 2005 – YouTube
- 2006 – Twitter
- 2007 – iPhone
- 2009 – WhatsApp





A BIT OF HISTORY

The 2010s

- NoSQL databases emerge as contenders to relational databases
- SQL databases start to incorporate NoSQL features
- The cloud makes enterprise grade features available to companies of all sizes

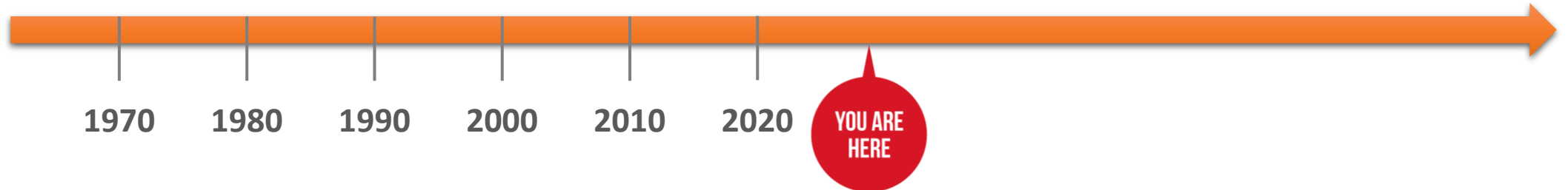




A BIT OF HISTORY

The 2020s

- Scale of data continues to grow
- AI on the rise
- SQL databases incorporate AI features
 - vector embeddings
 - AI driven self-tuning



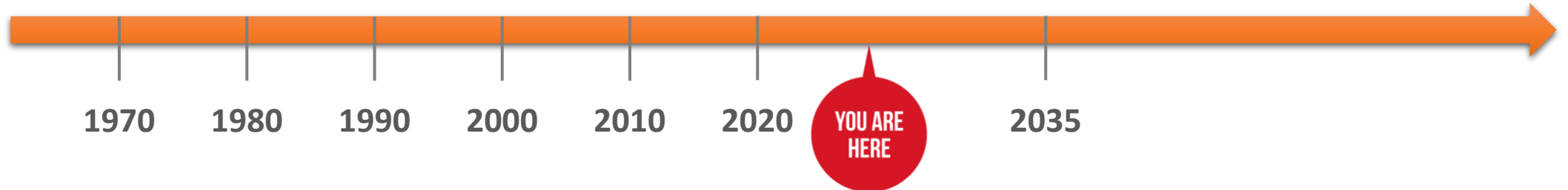


A BIT OF HISTORY

The Future

What are we looking for in a database:

- Cloud-native
- Seamless horizontal scaling
- AI integration
- Real-time analytics





CURRENT TRENDS

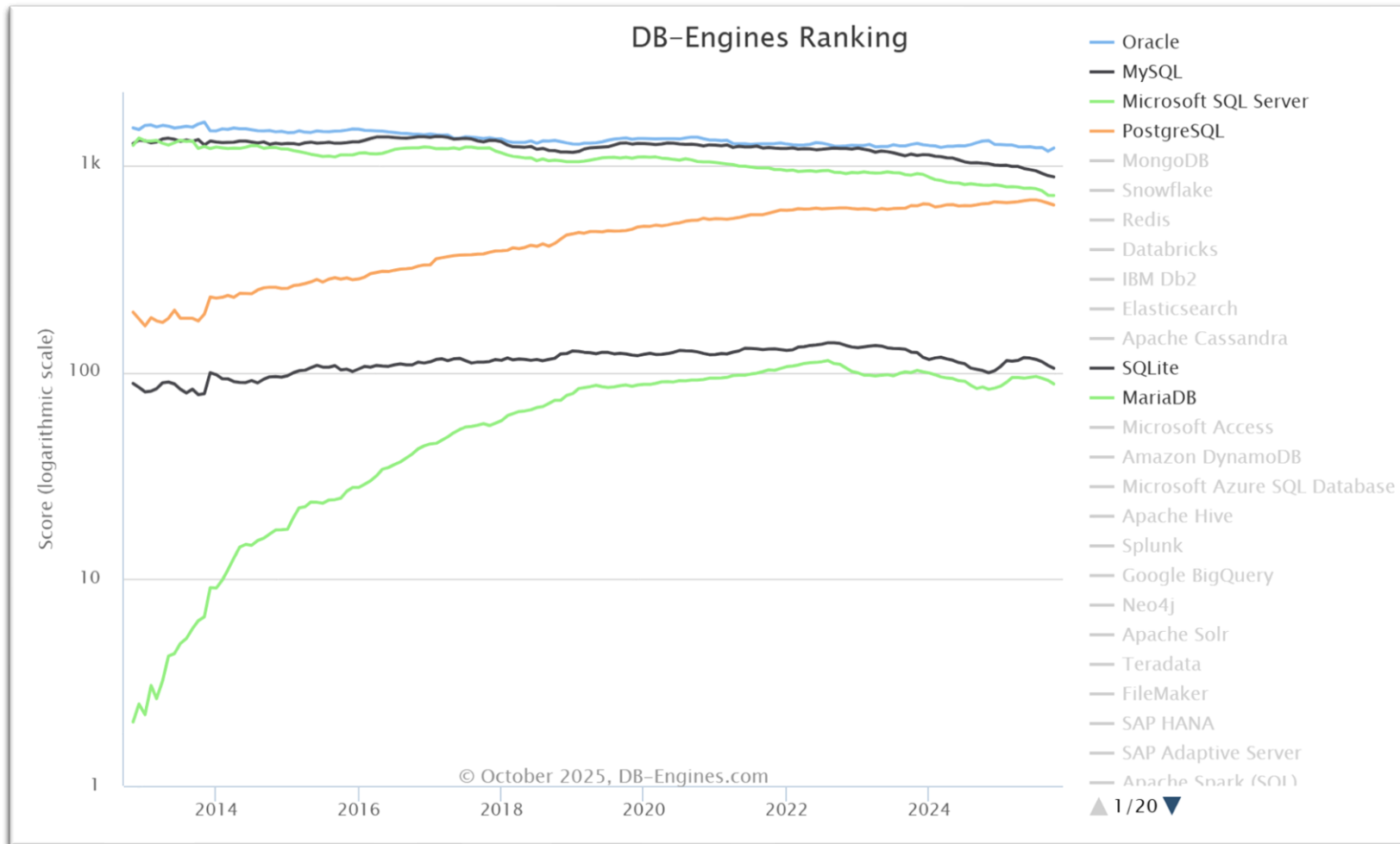
Is SQL Server's dominance declining?

Rank			DBMS	Database Model	Score		
Oct 2025	Sep 2025	Oct 2024			Oct 2025	Sep 2025	Oct 2024
1.	1.	1.	Oracle	Relational, Multi-model 	1212.77	+42.15	-96.67
2.	2.	2.	MySQL	Relational, Multi-model 	879.66	-12.11	-143.09
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model 	715.05	-2.27	-87.04
4.	4.	4.	PostgreSQL	Relational, Multi-model 	643.20	-13.97	-8.96
5.	5.	5.	MongoDB 	Document, Multi-model 	368.01	-12.49	-37.20
6.	6.	 7.	Snowflake	Relational	198.65	+8.46	+58.05
7.	7.	 6.	Redis	Key-value, Multi-model 	142.33	-2.84	-7.30
8.	 9.	 14.	Databricks	Multi-model 	128.80	+4.74	+43.21
9.	 8.	9.	IBM Db2	Relational, Multi-model 	122.37	-1.81	-0.40
10.	10.	 8.	Elasticsearch	Multi-model 	116.67	-1.59	-15.18



CURRENT TRENDS

Is SQL Server's dominance declining?



THE CHALLENGE





THE ELEPHANT IN THE ROOM

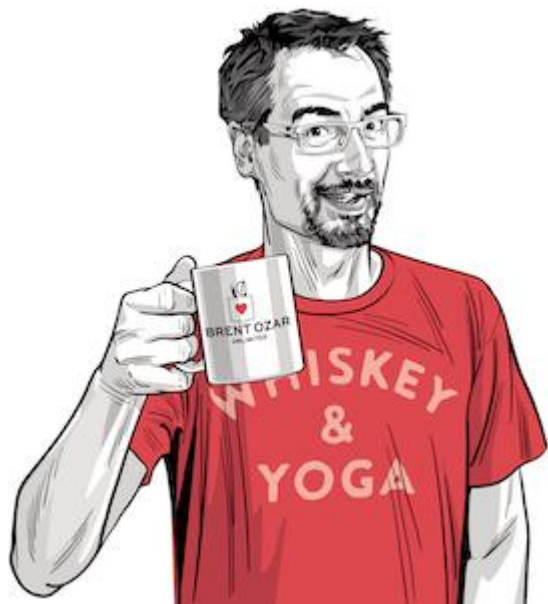
PostgreSQL's plans to rule the world

- Fastest growing open-source database
- DBMS of the year in 2023
- Offers enterprise-grade features
- Rich set of extensions
- Modern SQL feature (JSON, JSONB, RegExp...)
- Cloud-Native Ecosystem
- Vendor Neutrality
- ... will it rule the world?





FRIENDS TURN TO FOES



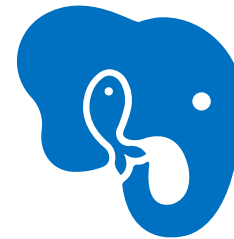




A DIRECT THREAT TO SQL SERVER

Babelfish promises drop-in replacement

- PostgreSQL extension
- Developed and maintained by AWS
- Open-source, active development
- Speaks T-SQL over TDS
- Offers low-friction migration path from SQL Server to PostgreSQL
- Provides migration tools and docs
- Not a silver bullet

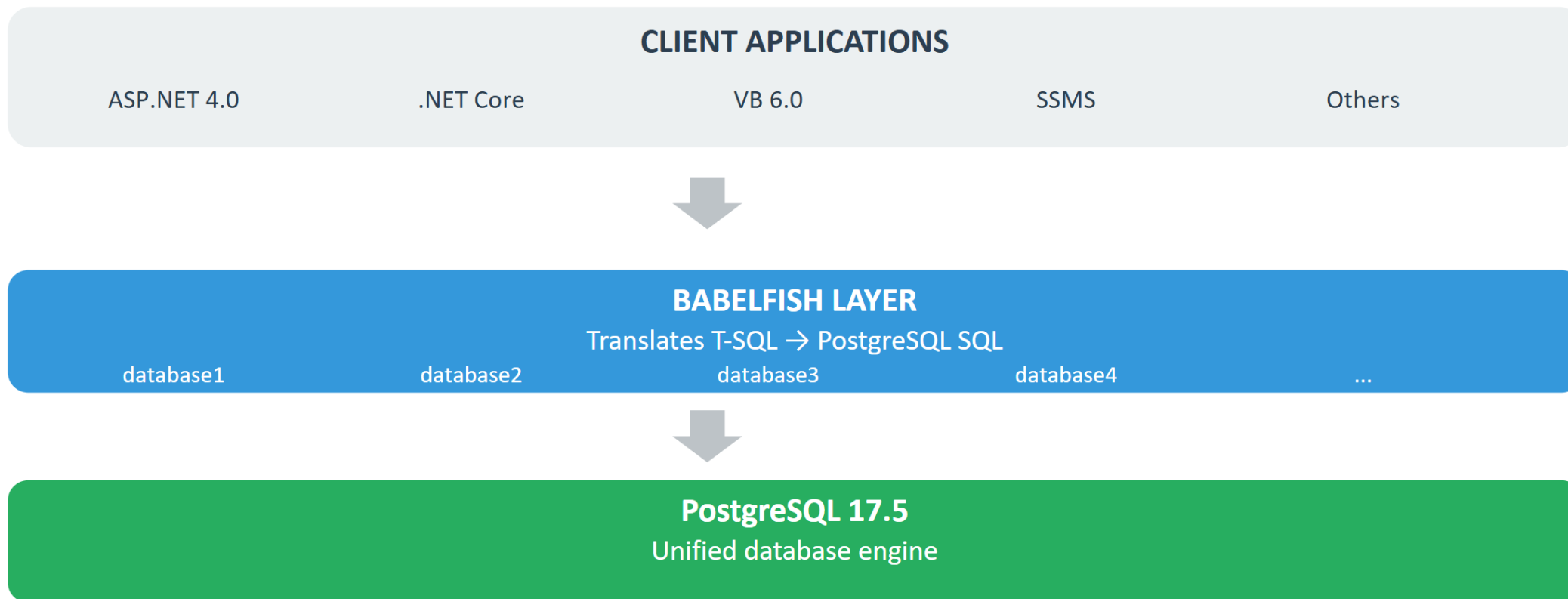


BABELFISH
for PostgreSQL



BABELFISH ABSTRACTION LAYER

You speak T-SQL, it speaks PGSQL





BABELFISH MULTI-DB MODE

Many databases in SQL Server, many schemas in PostgreSQL

T-SQL View (port 1433)

Database Structure:

Multiple Databases:

- database1 (with dbo, sales, hr)
- database2 (with dbo, sales, hr)
- database3 (with dbo, sales, hr)

Example Query:

```
USE database1;
SELECT * FROM dbo.customers;
SELECT * FROM sales.orders;
USE database2;
SELECT * FROM dbo.customers;
```

PostgreSQL View (port 5432)

Database Structure:

Database: babelfish_db (single!)

Schemas (prefixed with DB name):

- database1_dbo
- database2_dbo
- database1_sales
- database2_sales
- database1_hr
- database2_hr

Example Query:

```
SELECT * FROM database1_dbo.customers;
SELECT * FROM database1_sales.orders;
SELECT * FROM database2_dbo.customers;
```



BABELFISH SINGLE-DB MODE

One database in SQL Server, one database in PostgreSQL

T-SQL View (port 1433)

Database Structure:

Databases: MyDatabase

Schemas:

- dbo
- sales
- hr

Example Query:

```
USE database1;  
SELECT * FROM dbo.customers;  
SELECT * FROM sales.orders;  
SELECT * FROM hr.employees;
```

PostgreSQL View (port 5432)

Database Structure:

Database: babelfish_db

Schemas:

- dbo
- sales
- hr

Example Query:

```
SELECT * FROM dbo.customers;  
SELECT * FROM sales.orders;  
SELECT * FROM hr.employees;
```




ALL THAT GLITTERS IS GOLD?

Unsupported statements

ALTER DATABASE	CREATE, ALTER, DROP EXTERNAL LANGUAGE	CREATE TABLE... GRANT clause	SET FORCEPLAN
ALTER DATABASE SCOPED CONFIGURATION	CREATE, ALTER, DROP EXTERNAL RESOURCE POOL	CREATE TABLE... IDENTITY clause	SET NUMERIC_ROUNDABORT ON
ALTER DATABASE SCOPED CREDENTIAL	CREATE, ALTER, DROP FULLTEXT CATALOG	CREATE USER	SET OFFSETS
ALTER DATABASE SET HADR	CREATE, ALTER, DROP FULLTEXT INDEX	DENY	SET REMOTE_PROC_TRANSACTIONS
ALTER INDEX	CREATE, ALTER, DROP FULLTEXT STOPLIST	END, MOVE CONVERSATION	SET SHOWPLAN_TEXT
ALTER PARTITION FUNCTION	CREATE, ALTER, DROP MESSAGE TYPE	EXECUTE with AS LOGIN or AT option	SET SHOWPLAN_XML
ALTER PARTITION SCHEME	CREATE, ALTER, DROP, OPEN/CLOSE, BACKUP/RESTORE MASTER KEY	GET CONVERSATION GROUP	SET STATISTICS
ALTER SCHEMA	CREATE, ALTER, DROP QUEUE	GROUP BY ALL clause	SET STATISTICS PROFILE
ALTER SERVER CONFIGURATION	CREATE, ALTER, DROP RESOURCE GOVERNOR	GROUP BY CUBE clause	SET STATISTICS TIME
ALTER SERVICE, BACKUP/RESTORE SERVICE MASTER KEY clause	CREATE, ALTER, DROP RESOURCE POOL	GROUP BY ROLLUP clause	SET STATISTICS XML
BEGIN CONVERSATION TIMER	CREATE, ALTER, DROP ROUTE	INSERT... DEFAULT VALUES	SHUTDOWN statement
BEGIN DISTRIBUTED TRANSACTION	CREATE, ALTER, DROP SEARCH PROPERTY LIST	MERGE	UPDATE STATISTICS
BEGIN DIALOG CONVERSATION	CREATE, ALTER, DROP SECURITY POLICY	READTEXT	UPDATETEXT
BULK INSERT	CREATE, ALTER, DROP SELECTIVE XML INDEX clause	REVERT	Using EXECUTE to call a SQL function
CREATE COLUMNSTORE INDEX	CREATE, ALTER, DROP SERVICE	SELECT TOP x PERCENT WHERE x <> 100	VIEW... CHECK OPTION clause
CREATE EXTERNAL FILE FORMAT	CREATE, ALTER, DROP SPATIAL INDEX	SELECT TOP... WITH TIES	VIEW... VIEW_METADATA clause
CREATE EXTERNAL TABLE	CREATE, ALTER, DROP TYPE	SELECT... FOR BROWSE	WAITFOR DELAY
CREATE, ALTER, DROP APPLICATION ROLE	CREATE, ALTER, DROP XML INDEX	SELECT... FOR XML AUTO	WAITFOR TIME
CREATE, ALTER, DROP ASSEMBLY	CREATE, ALTER, DROP XML SCHEMA COLLECTION	SELECT... FOR XML EXPLICIT	WAITFOR, RECEIVE
CREATE, ALTER, DROP ASYMMETRIC KEY	CREATE/DROP RULE	SELECT... FOR XML PATH	WITH XMLNAMESPACES construct
CREATE, ALTER, DROP CREDENTIAL	CREATE, DROP WORKLOAD CLASSIFIER	SEND	WRITETEXT
CREATE, ALTER, DROP CRYPTOGRAPHIC PROVIDER	CREATE, ALTER, DROP WORKLOAD GROUP	SET DATEFORMAT	XPATH expressions
CREATE, ALTER, DROP ENDPOINT	ALTER TRIGGER	SET DEADLOCK_PRIORITY	
CREATE, ALTER, DROP EVENT SESSION		SET FMTONLY	

SQL SERVER FIGHTS BACK





WHAT MAKES SQL SERVER SPECIAL?

The challenge

Cost

Support

Development

Maintenance

Performance

Enterprise Features





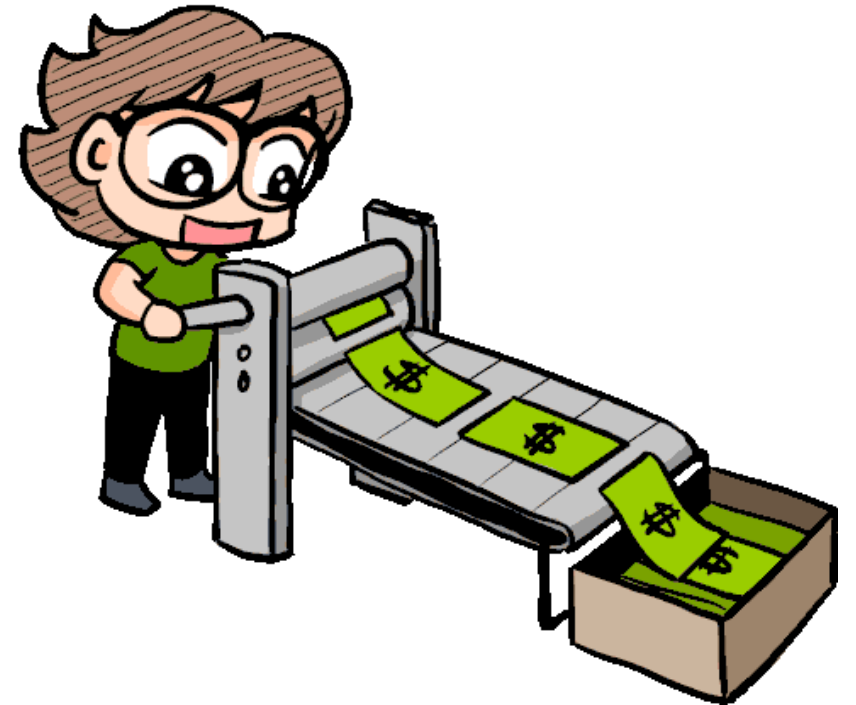
COST

Commercial vs. Open-Source



Analysis Services
Integration Services
Reporting Services
Master Data Services
Machine Learning Services

- SQL Server on-prem requires a license
 - Server + CAL € 1.000 per server + € 250 per CAL
 - Standard Edition € 2.000 per core
 - Enterprise Edition € 7.000 per core
- Oracle lives in a different universe
 - Standard Edition € 8.050 per processor
 - Enterprise Edition € 42.500 per processor
- PostgreSQL is free
 - Open source





CHEAP OUT ON FREE EDITION

Free as in beer

- SQL Server
 - Express Edition – 1 CPU + 1 GB RAM + ~~10 GB DB~~ 50 GB in 2025!!
 - Developer Edition – Unlimited for non-production use
 - Free tier on Azure SQL
- Oracle
 - Express Edition – 2 CPU + 2 GB RAM + 20 GB DB
 - Personal Edition – not free but inexpensive for dev use
 - Oracle Cloud Free Tier
- PostgreSQL
 - Everything is for free



THE “OPEN-SOURCE TAX”

- Lack of official support
 - Community-centric support – AKA “RTFM”
- Sparse or outdated documentation
- Dependency hell
 - “Works on my machine” syndrome
- DIY Troubleshooting
 - StackOverflow, Reddit, GitHub issues, Source Code
- Mitigated by the cloud
 - All is taken care by pros





EVERY CLOUD HAS A SILVER LINING

... but can cost you serious money

Platform	Engine	vCPU	RAM	Storage	DB Cost/ month	Storage/ month	License included?	Main notes
AWS RDS for SQL Server Standard	SQL Server	4	16 GB	128 GB	~\$839	~\$15	Yes	MS support, higher cost
AWS RDS for PostgreSQL	PostgreSQL	4	16 GB	128 GB	~\$532	~\$23	Yes (open-source)	Much cheaper
Azure SQL Managed Instance (GP)	SQL Server	4	16 GB	128 GB	~\$800	~\$11	Yes	Enterprise features, MS support
Azure Database for PostgreSQL Flex.	PostgreSQL	4	16 GB	128 GB	~\$260	~\$14	Yes (open-source)	Replica, HA, Azure integration



SUPPORT

SQL Server

- Official Microsoft Support
 - Several options available
- Cloud Support
- Outstanding documentation
- Partners and consultants
- Legacy product support
 - Mainstream support
 - Extended support
- Community

PostgreSQL

- No official support channel
 - EnterpriseDB (EDB) is the leader
- Cloud Support
- Good documentation
- Partners and consultants
- No legacy support
 - Partners
 - Community
- Community



COMMUNITY

Sharing is caring

SQL Server

- Very Active
- MVPs, bloggers, speakers
- International and regional events
- Forums and support channels
- Sharing culture, friendly attitude

PostgreSQL

- Very Active
- Very technical
- International and regional events
- Forums and mailing lists
- Rigorous but unfriendly attitude
 - “RTFM” prone answers



DEVELOPMENT TOOLS

SQL Server

- Visual Studio Database Projects
 - Versioning
 - Schema compare
 - Refactoring
 - Automated deployment
 - Unit testing
 - CI/CD integration
 - Top notch IDE experience

PostgreSQL

- Nothing like Visual Studio
 - pgAdmin ERD Tool
 - pgModeler
- Liquibase/Flyway for CI/CD
- Command-line tools



DEVELOPMENT

A true Swiss Army Knife

PostgreSQL has unique development features

- Countless supported **languages** (SQL, pgSQL, python, java, PHP, R, Lua, Rust, Ruby, Perl...)
- Native types, **table types** (with inheritance!)
- **JSON/JSONB** native data type for NoSQL features
- Fulltext search, RegExp, advanced text manipulation
- **Several extensions***
 - TimescaleDB
 - pgVector
 - Apache AGE (graph)
 - pgML
 - postgREST
 - pgTap unit testing
 - pg_graphql
 - omnigres (containers)
 - pg_doom

***open-source tax applies**





MANAGEMENT TOOLS

SQL Server

- Management Studio (SSMS)
- Visual Studio Code Extension
- Sqlcmd
- More “visual”

PostgreSQL

- pgAdmin
- Dbeaver / DataGrip / HeidiSQL
- psql
- More “scripting”

Scripts are good for automation

GUI helps with troubleshooting, especially with inexperienced DBAs
or when an outage hits



PERFORMANCE

Indexing

SQL Server

- Columnstore indexes
- Filtered Indexes / Indexed Views
- In-memory OLTP
- Parallelism
- Advanced Partitioning

PostgreSQL

- GIN (General Inverted Indexes)
- GiST (Generalized Search Tree)
- Hash indexes
- Partitioning
- Columnstore and in-mem available with extensions



PERFORMANCE

Performance tuning

SQL Server superior performance tuning tools

- Rich query execution plans
- Live execution plans
- Query Store
- Extended Events
- Automatic tuning
- Countless DMVs to observe every possible entity



ENTREPRISE FEATURES

Security

Feature	SQL Server	PostgreSQL
Encryption at rest	Transparent Data Encryption (TDE)	pgcrypto (extension), or via filesystem/disk
Row-level security	Yes, native	Yes, native (since v9.5+)
Data masking	Yes, native	Plugin/extension (e.g. pg_masking)
Auditing	Yes, very advanced	Basic (log statement), external plugins (pgaudit)
Always Encrypted	Yes, native (client-side encryption)	Needs external tool or custom implementation



ENTERPRISE FEATURES

Compression & Storage

Feature	SQL Server	PostgreSQL
Table/Index compression	Row/Page/Columnstore compression, native	TOAST (automatic), extensions for compression
Columnstore index	Yes, native, very advanced	Extensions (e.g. cstore_fdw), less integrated
Partitioning	Yes, native, advanced	Yes, native (since v10)
FileStream/FileTable	Yes, manage BLOBs/files in DB or filesystem	BLOBs via bytea or Large Object, less flexible



ENTERPRISE FEATURES

High Availability and Disaster Recovery

Feature	SQL Server	PostgreSQL
High Availability	Always On AG, FCI, log shipping, mirroring	Streaming replication, Patroni, repmgr
Advanced backup/restore	Yes, integrated, point-in-time, offloaded backup	Yes, via CLI tools (pg_dump, basebackup)
Geo-replica	Yes, native	Yes, complex configuration
Online restore	Yes, partial (page/partial restore)	More limited, often full restore

- PostgreSQL lacks automatic failover and automated client routing
- PostgreSQL requires external tools and/or extensions:
 - Patroni + etcd/consul/ZooKeeper
 - haproxy
 - repmgr
 - pgBouncer



ENTERPRISE FEATURES

Other features

Feature	SQL Server	PostgreSQL
Change Data Capture (CDC)	Yes, native	Extension (pgoutput, wal2json, Debezium)
Temporal tables	Yes, native	Extension (temporal_tables)
Data masking	Yes, native	Extension
Service Broker (messaging)	Yes, native	External plugins (PGQ, others)
JSON support	Yes	Yes, very advanced
GIS/Spatial	Yes, native (SQL Server Spatial)	Yes, PostGIS (very advanced)

CONCLUSIONS





KEEP THE FAITH

- SQL Server is still the winner for Enterprise Features
- PostgreSQL provides many features as extensions
- The cloud reduces friction and “open-source tax”
- Development and Maintenance are better in SQL Server
- PostgreSQL offers more possibilities, requires more tinkering

Do not attempt a migration!

Tools like Babelfish make it look easy. It's a trap.



IS SQL SERVER STILL MY FAVORITE CHILD?





FEEDBACK PLEASE!





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Thanks!

